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ORIGINAL ARTICLE

Genome-wide characterization of *phenylalanine ammonia-lyase* gene family in watermelon (*Citrullus lanatus*)

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Abstract Phenylalanine ammonia-lyase (PAL), the first enzyme in the phenylpropanoid pathway, plays a critical role in plant growth, development, and adaptation. PAL enzymes are encoded by a gene family in plants. Here, we report a genome-wide search for *PAL* genes in watermelon. A total of 12 *PAL* genes, designated *ClPAL1-12,* are identified . Nine are arranged in tandem in two duplication blocks located on chromosomes 4 and 7, and the other three *ClPAL* genes are distributed as single copies on chromosomes 2, 3, and 8. Both the cDNA and protein sequences of *ClPAL*s share an overall high identity with each other. A phylogenetic analysis places 11 of the ClPALs into a separate cucurbit subclade, whereas ClPAL2, which belongs to neither monocots nor dicots, may serve as an ancestral PAL in plants. In the cucurbit subclade, seven ClPALs form homologous pairs with their counterparts from cucumber. Expression profiling reveals that 11 of the *ClPAL* genes are expressed and show preferential expression in the stems and male and female flowers. Six of the 12 *ClPAL*s are moderately or strongly expressed in the fruits, particularly in the pulp, suggesting the potential roles of PAL in the development of fruit color and flavor. A promoter motif analysis of the *ClPAL* genes implies redundant but distinctive *cis*-regulatory structures for stress responsiveness. Finally, duplication events during the evolution and expansion of the *ClPAL* gene family are discussed, and the relationships between the *ClPAL* genes and their cucumber orthologs are estimated.

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Abbreviations

ABA	Abscisic acid
ABRE	ABA-responsive element
AuxRE	Auxin-responsive element
CRT	C-repeat
ERE	Ethylene-responsive element
GA	Gibberellin
GARE	GA-responsive element
HSE	Heat shock-responsive element
LTRE	Low temperature-responsive element
MIO	4-Methylidene-imidazolone-5-one
ORF	Open reading frame
PAL	Phenylalanine ammonia-lyase
RT-PCR	Reverse transcription-polymerase chain reaction
SA	Salicylic acid
SARE	SA-responsive element
UTR	Untranslated region
WGD	Whole-genome duplication

Introduction

The phenylpropanoid pathway serves as a rich source of plant secondary metabolites, which are precursors of a wide range of compounds with various functions in plants (Vogt 2010). Monolignols, the products of the phenylpropanoid pathway, can function as the building blocks of lignin, which confers structural support and vascular integrity to plants and, therefore, is essential for stem rigidity and for conducting water, minerals, and photosynthetic products (Ferrer et al. 2008). Flavonoids, isoflavonoids, coumarins, and stilbenes, all of which are synthesized from phenylpropanoid compounds, have important functions in plant defenses against pathogens, as UV light protectants, or as regulatory molecules in signal transduction and communication with other organisms (Ferrer et al. 2008).

Phenylalanine ammonia-lyase (PAL; EC 4.3.1.5) is the first enzyme in the phenylpropanoid pathway. PAL catalyzes the non-oxidative deamination of phenylalanine to *trans*-cinnamate, a reaction that is generally considered to be a key regulatory point between primary and secondary metabolism (Fraser and Chapple 2011). In all studied plants, PAL is encoded by a multi-gene family. The number of *PAL* gene copies is four in Arabidopsis (Raes et al. 2003) and tobacco (Fukasawa-Akada et al. 1996; Reichert et al. 2009), five in poplar (Hamberger et al. 2007), seven in cucumber (Shang et al. 2012), and nine in rice (Hamberger et al. 2007). The individual *PAL* gene may respond differentially to biotic or abiotic stress, and their expression is developmentally and spatially controlled. In poplar, the organ-specific expression of *PAL* genes has been assayed, with two poplar *PAL* genes being expressed in lignifying tissues: one involved in lignin formation and the other one specially targeted to condensed tannin formation (Kao et al. 2002). A third poplar *PAL* gene is associated with flowering, though its functional roles remain to be established (Hamberger et al. 2007). In Arabidopsis, expression studies of *AtPAL*s have shown that *AtPAL3* is expressed only at basal levels in stems (Mizutani et al. 1997; Raes et al. 2003). In contrast, *AtPAL1*, *AtPAL2*, and *AtPAL4* are expressed at relatively high levels in stems during the late developmental stages, whereas *AtPAL2* and *AtPAL4* are both expressed in seeds, and *AtPAL1* expression has been localized to vascular tissue (Raes et al. 2003; Rohde et al. 2004). Of the four *AtPAL*s, only *AtPAL1* and *AtPAL2* are induced in response to decreased nitrogen and low temperatures (Olsen et al. 2008). In contrast to Arabidopsis and poplar, a remarkable set of ~26 *PAL* genes have been detected in tomato; however, only a single gene appears to be strongly expressed in all tissues, whereas the remaining *PAL*s appear to be effectively silenced (Chang et al. 2008).

The expansion of *PAL* gene families in plants is likely due to gene duplication, including tandem duplication, segmental duplication, and whole-genome duplication (Cannon et al. 2004). For example, seven *PAL* genes are arranged in tandem in two duplication blocks in cucumber, suggesting that the *PAL* gene family may have expanded in the cucumber genome through tandem duplications (Shang et al. 2012). In contrast, more than 20 *PAL* genes are widely dispersed in the tomato genome, with no clustering, indicating an unusually active duplication event (Chang et al. 2008).

Watermelon (*Citrullus lanatus*) belongs to the cucurbit family, which includes several other important vegetable crops, such as cucumber, melon, and pumpkin. According to statistics from Food and Agriculture Organization (FAO), watermelon accounts for approximately 7 % of the agricultural area devoted to vegetable crops (http://faostat.fao.org). In addition, the sequencing of the watermelon genome (Guo et al. 2013) affords a unique opportunity to identify all members of *PAL* family in watermelon. In this study, we present a detailed analysis of the phylogeny, synteny, and gene structure of the watermelon *PAL* family and their tandem arrangement, expression profiles, and regulatory *cis*-elements. In particular, a comparison of the *PAL* families between watermelon and cucumber provides further evidence for the evolution of *PAL* gene families in cucurbits.

Materials and methods

Plant growth conditions

Watermelon (*C. lanatus* subsp. *vulgaris* cv. Jingxin No. 1) seeds were germinated and grown in a greenhouse in pots containing a steam-sterilized soil mix (peat moss:perlite:vermiculite, 3:1:1, v/v/v). The growth conditions were as follows: a 12-h photoperiod, temperature of 28/18 °C (day/night), and light intensity of 600 μmol m^{-2} s^{-1}. Seedlings at the two-leaf stage were harvested for tissues (roots, stems, leaves, and cotyledons); for flower samples, male and female flowers were harvested at 2 days after opening. Fruit samples were harvested at the fully red stage (~35 dpa). All tissue samples were immediately frozen in liquid nitrogen and stored at −80 °C until use.

Database searching and sequence analysis

The *ClPAL* sequence data used in this study were collected by homology screening against "Watermelon Genome Database" (IWGI, http://www.iwgi.org/watermelon.html; Ren et al. 2012). Homology searches were performed using tBLASTn with the default parameters, and the known PAL sequences from Arabidopsis and cucumber were used as queries.

Intron delimitation of the target genes was performed by comparing the cDNA sequence with its genomic sequence derived from the IWGI database. The molecular weights and isoelectric points of the deduced proteins were predicted by ExPASy (http://www.cn.expasy.org/tools).

Approximately 1,500–2,000 bp of upstream sequence of the translation initiation codon ATG was chosen and isolated as the regulatory promoter region. These sequences were then subjected to PLACE (http://www.dna.affrc.go.jp/PLACE/) and PlantCARE database (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) analyses for *cis*-elements.

Phylogenetic analysis

PAL amino acid sequences were downloaded from the GenBank and ICuGI databases (http://www.icugi.org/cgi-bin/ICuGI/index.cgi) and aligned using Clustal X (version 1.81) (Thompson et al. 1997). A phylogenetic tree was then constructed using the MEGA software (version 5.0) and the Maximum-Likelihood (ML) method with 100 bootstrap replicates. The following PAL amino acid sequences were used for the phylogenetic analysis: AtPAL1 (NP_181241), AtPAL2 (NP_190894), AtPAL3 (NP_196043), and AtPAL4 (NP_187645) from *Arabidopsis thaliana*; NtPAL1 (AAA34 122), NtPAL2 (ACJ66298), NtPAL3 (BAA22963), and NtPAL4 (CAA55075) from *Nicotiana tabacum*; OsPAL1 (BAD23149), OsPAL2 (BAD23155), OsPAL3 (CBC40318), and OsPAL4 (CBC40320) from *Oryza sativa*; ZmPAL1 (AAL40137), ZmPAL2 (NP_001147922), ZmPAL3 (NP_00 1147433), and ZmPAL4 (NP_001151482) from *Zea mays*; VvPAL (XP_002278516) from *Vitis vinifera*; PtPAL (AAA84889) and PpPAL (AAT66434) from *Pinus taeda* and *P. pinaster*, respectively; and CsPAL1 (Csa002863), CsPAL2 (Csa002864), CsPAL3 (Csa011894), CsPAL4 (Csa011895), CsPAL5 (Csa011896), CsPAL6 (Csa011940), and CsPAL7 (Csa011941) from *Cucumis sativus*, as indicated previously (Shang et al. 2012).

Semi-quantitative reverse transcriptase-polymerase chain reaction (RT-PCR)

The total RNA of watermelon seedlings was extracted from frozen tissue using TRIzol Reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's protocol. A total of 2 μg of RNA was used as the template for the first-strand cDNA synthesis using an RNA PCR kit (AMV, version 3.0, TaKaRa, Shuzo, Otsu, Japan). The PCR was performed using gene-specific primers (Supplementary Table S1). A housekeeping gene, the *18S rRNA* gene (HQ025803), was amplified as an internal control. The amplified products were separated on a 1.0 % agarose gel for quantitative analysis.

Results

Identification of *ClPAL* genes in the watermelon genome

To identify the complete *PAL* gene family in watermelon, all of the chromosomes and scaffold sequences in the Watermelon Genome Database (IWGI) were searched using the PAL proteins from Arabidopsis (AtPALs) and cucumber (CsPALs) as queries. A total of 12 *PAL* genes were identified (Table 1), and these watermelon *PAL* genes were mapped to five chromosomes. We renamed them *ClPAL1* to *ClPAL12* based on their order on the chromosomes, from chromosome 2 to 8 (Table 1).

Nine of the 12 *ClPAL* genes were distributed as two clusters (Table 1). The first cluster possesses seven *ClPAL* genes (*ClPAL3-9*), which are arranged in tandem on chromosome 4 (Table 1). In this cluster, *ClPAL3-5* is in the forward direction, whereas the others are oriented in the reverse direction (Table 1). The other cluster is located on chromosome 7 and contains *ClPAL10* and *ClPAL11*, with their open reading frames (ORFs) oriented in opposite directions and separated by 1,709 bp (Table 1). However, the other three *ClPAL* genes (*ClPAL1*, *ClPAL2*, and *ClPAL12*) are located on different chromosomes as single copies (Table 1).

Next, to establish whether these *ClPAL* genes were expressed, we cloned and sequenced the full-length genomic DNA and cDNA of *ClPAL1-12*. The gene annotations for these loci were manually verified and improved; the annotated nucleotide sequences and the resulting amino acid sequences of *ClPAL1-12* are presented in the Supplementary Fig. S1. The lengths of the *ClPAL* ORFs were slightly different, ranging from 2,109 to 2,151 bp (Table 1). These ORFs encoded polypeptides of approximately 710 amino acids, with a predicted molecular mass and theoretical p*I* of approximately 78 kDa and 6.0, respectively (Table 1), which were consistent with the molecular mass and p*I* values of the PAL proteins from other plants (Kumar and Ellis 2001; Reichert et al. 2009; Shang et al. 2012).

Exon–intron organization of the *ClPAL* gene family

Analysis of the exon–intron structure of the *ClPAL* genes revealed some variations. Nine of the *ClPAL* genes (*ClPAL2*, *ClAPL4-11*) contained no introns in their ORFs (Fig. 1a), agreeing with the the previous findings , where no intron was present in any of the seven *CsPAL* genes (Shang et al. 2012). However, the ORFs of the other three *ClPAL* genes were interrupted by a single intron (Fig. 1a), the length of which varied substantially, ranging from 871 bp in *ClPAL3* to more than 2,100 bp in *ClPAL1* and *ClPAL12* (Fig. 1a). A similar exon–intron organization was also detected in known *PAL* genes, including Arabidopsis *PAL1* and *PAL2,* and *NtPAL1* from tobacco; the other Arabidopsis *PAL* genes (*AtPAL3* and *AtPAL4*) contain one additional intron (Fig. 1b). Of note, the length of exon 2 was highly conserved (1,750 bp) in all the plant *PAL* genes with one intron (Fig. 1).

Table 1 *ClPAL* genes and properties of the deduced proteins

Gene	IWGI no.	Chromosome	Position (start)	Position (end)	Orientation	ORF length (bp)	Protein		
							Size (aa)	MW (Da)	p*I*
ClPAL1	*Cla008727*	2	31,544,760	31,549,081	Forward	2,142	713	77,490.52	6.31
ClPAL2	*Cla011180*	3	26,014,434	26,016,569	Forward	2,136	711	78,069.25	5.63
ClPAL3	*Cla018297*	4	20,607,783	20,610,798	Forward	2,145	714	78,082.08	6.00
ClPAL4	*Cla018298*	4	20,617,081	20,619,213	Forward	2,133	710	77,434.23	5.87
ClPAL5	*Cla018299*	4	20,625,584	20,627,596	Forward	2,127	708	77,512.29	5.74
ClPAL6	*Cla018300*	4	20,632,484	20,630,343	Reverse	2,142	713	78,146.13	6.06
ClPAL7	*Cla018301*	4	20,637,440	20,635,290	Reverse	2,151	716	78,412.58	6.31
ClPAL8	*Cla018302*	4	20,641,642	20,639,534	Reverse	2,109	702	76,665.72	6.32
ClPAL9	*Cla018303*	4	20,646,557	20,644,440	Reverse	2,118	705	77,701.06	5.99
ClPAL10	*Cla012779*	7	26,423,015	26,420,874	Reverse	2,142	713	77,851.70	5.99
ClPAL11	*Cla012780*	7	26,424,725	26,426,854	Forward	2,130	709	77,404.09	5.96
ClPAL12	*Cla013761*	8	16,922,833	16,927,084	Forward	2,142	713	78,351.44	6.17

The theoretical molecular weight (Mw) and isoelectric point (p*I*) were calculated by ExPASy (http://cn.expasy.org/tools). Nucleic acid and deduced amino acid sequences of each *ClPAL* gene are listed in Supplementary Fig. S1

A close analysis of the intron/exon boundaries of these *PAL* genes, both the nucleic acid and amino acid sequences around the splice sites, exhibited high homology (Fig. 2). The intron is inserted between the second and third bases of a conserved Arg codon at a position that is identical to that observed in other angiosperm *PAL* genes (Fig. 2). Additionally, this intron insertion site is conserved in *AtPAL3* and *AtPAL4*, whose coding regions are interrupted by two introns (data not shown). Furthermore, the intron/exon junctions conform to the eukaryotic consensus GT/AG, but not to the AT/AC rule (Mount 1996) for splice donors and acceptors (Fig. 2).

Sequence characterization of *ClPAL* genes

The *ClPAL* genes showed high identity to each other at both the nucleotide and protein levels, though *ClPAL2* shared only approximately 60 % identity with the others (Supplementary Table S2 and S3). Among the 12 *ClPAL* genes, the highest nucleotide sequence identity in the coding region occurred between *ClPAL10* and *ClPAL11* (99.0 %, resulting in 98.2 % identity at the protein level).

The examination of the ClPAL protein sequences identified four functional domains: an N-terminal region, an MIO (4-methylidene-imidazolone-5-one) domain, a core domain, and the inserted shielding domain (Fig. 3a), as noted previously in other plant PAL proteins (Ritter and Schulz 2004). A detailed protein sequence alignment of ClPALs is shown in Fig. 3b. The greatest divergence occurred in the N-terminal region, as also described for the PAL families in bean (Cramer et al. 1989), raspberry (Kumar and Ellis 2001), and cucumber (Shang et al. 2012).

As indicated in Fig. 3b, the conserved enzymatic active site Ala-Ser-Gly is found in the MIO domains of all the ClPAL proteins. The single intron insertion site of *ClPAL1*, *ClPAL3*, and *ClPAL12* was also located in the MIO domain (Fig. 3b). In the core domain, two conserved residues, Tyr340 and Gly480 (numbering according to ClPAL8), were found to be conserved in all the ClPAL proteins. Tyr340 is involved in proton release, and Gly480 is located at the base of active site pocket (Ritter and Schulz 2004); their mutation inactivates the PAL enzyme (Reichert et al. 2009). Another important residue is Thr535, which is present in the shielding domain and can be phosphorylated. This conserved Thr residue in the PAL proteins from bean, raspberry, and cucumber (Kumar and Ellis 2001; Shang et al. 2012) has suggested that phosphorylation may be a ubiquitous regulatory mechanism in higher plants. PAL phosphorylation might be associated with a decrease in *Vmax*, in agreement with the suggestion that protein phosphorylation is involved in tagging PAL subunits for turnover (Allwood et al. 1999). Although this phosphorylation site is conserved in nine of ClPALs, the Thr residue has been substituted by Tyr, Ile, and Val in ClPAL2, ClPAL9, and ClPAL12, respectively (Fig. 3b). All of these results suggested that all of the ClPAL proteins might be enzymatically active but that their activities are divergent.

To determine the evolutionary relationship among the plant PAL proteins, sequences of a set of PAL family members from other plant species, including cucumber, Arabidopsis, tobacco, grape, rice, and maize, were analyzed using a maximum-likelihood (ML) phylogenetic tree. The PAL proteins from gymnosperm plants, *Pinus taeda* and *Pinus pinaster*, were also included. As shown in Fig. 4, the corresponding tree divided these PALs into four clades of dicot, monocot, gymnosperm, and others. As expected, 11 of the ClPAL proteins were classified as members of the dicot group. The only one exception was ClPAL2, which was classified into neither the dicot nor monocot clade but clustered more closely with the PALs from gymnosperm plants (Fig. 4). In the dicot clade, the plant PALs within one species were more closely related to each other than to the homologs in the other species. However, the PALs from watermelon and cucumber were clustered together and formed a cucurbit subclade (Fig. 4). Based on this finding, it could be concluded that the diversification of plant PALs might have occurred separately in each species but was shared between watermelon and cucumber.

More interestingly, for the ClPAL4-8 proteins in the cucurbit subclade, each clustered with a single cucumber PAL protein (Fig. 4), indicating that *PAL* gene duplication arose prior to the split of watermelon and cucumber. However, the ClPAL10/ClPAL11 pair clustered with the CsPAL1/CsPAL2 pair (Fig. 4), placing this local duplication event after the split of watermelon and cucumber.

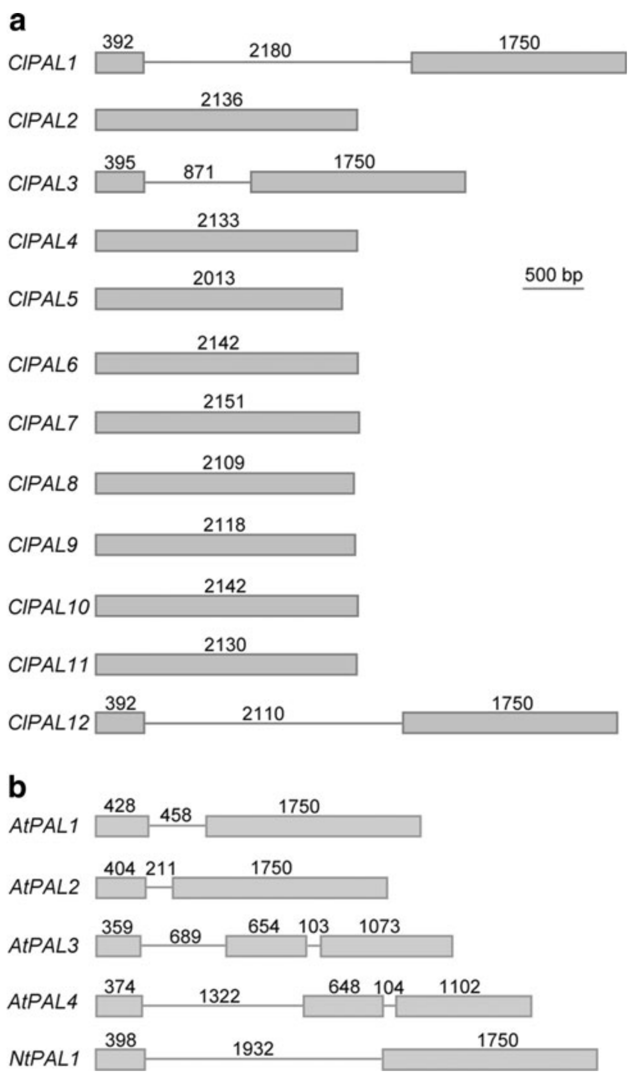


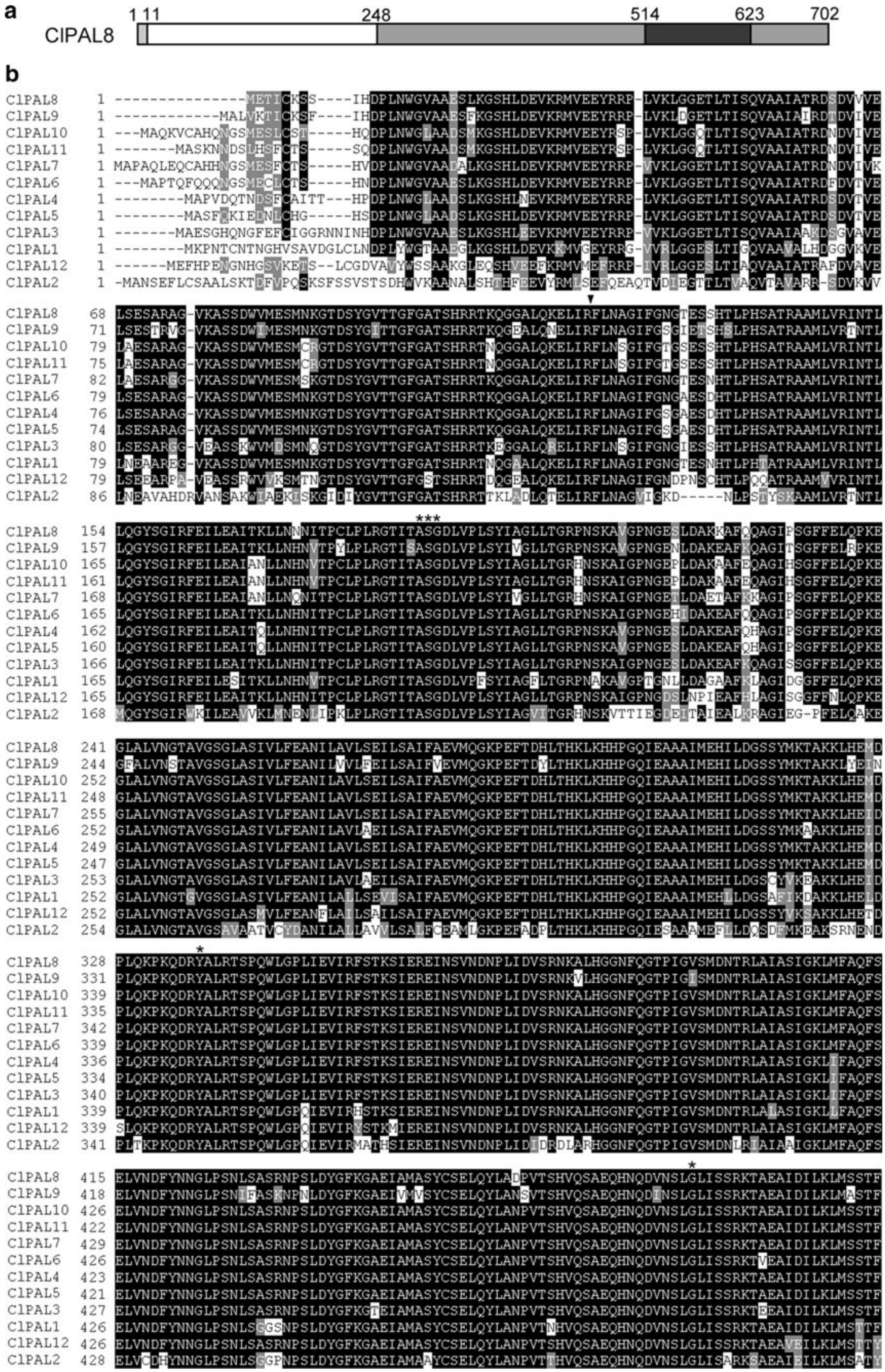
Fig. 1 Schematic diagrams of the exon–intron structures of *ClAPL* genes (**a**) and *PAL* genes from Arabidopsis and tobacco (**b**). The *boxes* and *lines indicate* the exons and introns, respectively, and their *lengths* are indicated in base pairs

Expression profiles of *ClPAL* genes in watermelon

To elucidate the physiological functions of different members of the PAL family in watermelon, the expression of *ClPAL* genes was investigated in different tissues, including the roots, stems, cotyledons, leaves, male and female flowers, and fruits. Because of the high similarity among the mRNAs of *ClPAL* genes and their nearly identical transcript sizes, we could not distinguish among them using real-time RT-PCR. Thus, we performed semi-quantitative RT-PCR using specific primers for each *ClPAL* gene (Supplementary Table S1). The proper amplification of the corresponding gene was confirmed by DNA sequencing. The watermelon 18S ribosomal RNA (*18S rRNA*) gene, which was stably expressed in different tissues (Li et al. 2012), was used as the internal control. The primer specificity and expression stability of *18S rRNA* were validated by an additional real-time PCR. The melting curve analysis of the PCR product confirmed that the primers amplified a single product, and the C_t values (18.61 ± 0.43) suggested a constant expression of *18S rRNA* among all tested tissues (Supplementary Fig. S3).

As shown in Fig. 5, all of the *ClPAL* genes, except *ClPAL2*, exhibited a rather broad expression in at least one of the tested tissues; although *ClPAL2* did not show any

Fig. 2 The nucleotide sequence alignment of the exon 1/intron/exon 2 region in the *ClPAL*, *AtPAL*, and *NtPAL* genes. The intron sequences are in lowercase letters


a
1 11
248
514
623
702
ClPAL8
b

◀**Fig. 3** Sequence analysis of ClPAL proteins. **a** Schematic diagram of the ClPAL amino acid sequence. ClPAL8 is shown as a representative of the ClPAL protein family. The four functional domains, including the N-terminal domain (*light gray*), MIO domain (*white*), core domain (*dark gray*), and inserting shielding domain (*black*), are indicated. **b** Alignment of multiple ClPAL protein sequences. The alignment was performed using Clustal X, followed by shading with Boxshade 3.21. The conserved residues are marked with *asterisk*, and the intron insertion site is marked by a *triangle*. The gaps are indicated as *dashes*

detectable expression in the tested tissues (Fig. 5), it may be expressed in other tissues, e.g., seeds, or its expression may be induced by various environmental stresses. For the other *ClPAL* genes, the amount of expression varied in the different tissues. *ClPAL1* was expressed at high levels in the roots, stems, and flowers but at relatively low levels in the cotyledons and leaves. In fruits, high *ClPAL1* expression was detected in the pulp, but only trace expression was found in the peel (Fig. 5). *ClPAL3* showed a similar expression profile with *ClPAL1*, in addition to lower expression levels in male and female flowers. *ClPAL4* and *ClPAL5* showed similar expression patterns: both transcripts could be detected in every reproductive tissue investigated, though for vegetative tissues, high expression was only detected in the stems (Fig. 5). *ClPAL6* was expressed at a relatively high level in the stems and male and female flowers, and the expression of *ClPAL6* was relatively lower in the roots and undetectable in leaves and fruits (Fig. 5). Additionally, the levels of *ClPAL7* mRNA were notably high in the roots and flowers, but barely detected in the cotyledons, leaves, and fruits. In contrast, a constitutive expression pattern was found for *ClPAL8* and *ClPAL9* that was higher in vegetative tissues (except in the cotyledons) than in reproductive tissues. Furthermore, the expression of *ClPAL9* was highest among all of the *ClPAL* genes, indicating the potentially important roles of ClPAL9 in watermelon growth and development. In addition, a similar expression pattern was also observed between *ClPAL10* and *ClPAL11*, which were principally expressed in the roots and stems, though their expression levels in the other tissues were very low or barely detectable (Fig. 5). The expression profiles for *ClPAl12* differed from the other *ClPAL* genes, with most of the *ClPAL12* transcripts accumulating in mature leaves (the second leaves, L2) and male flowers (Fig. 5). Overall, the overlapped but diverse expression patterns of the *ClPAL* genes suggested that ClPAL isoforms may play redundant but different roles at different developmental stages in watermelon.

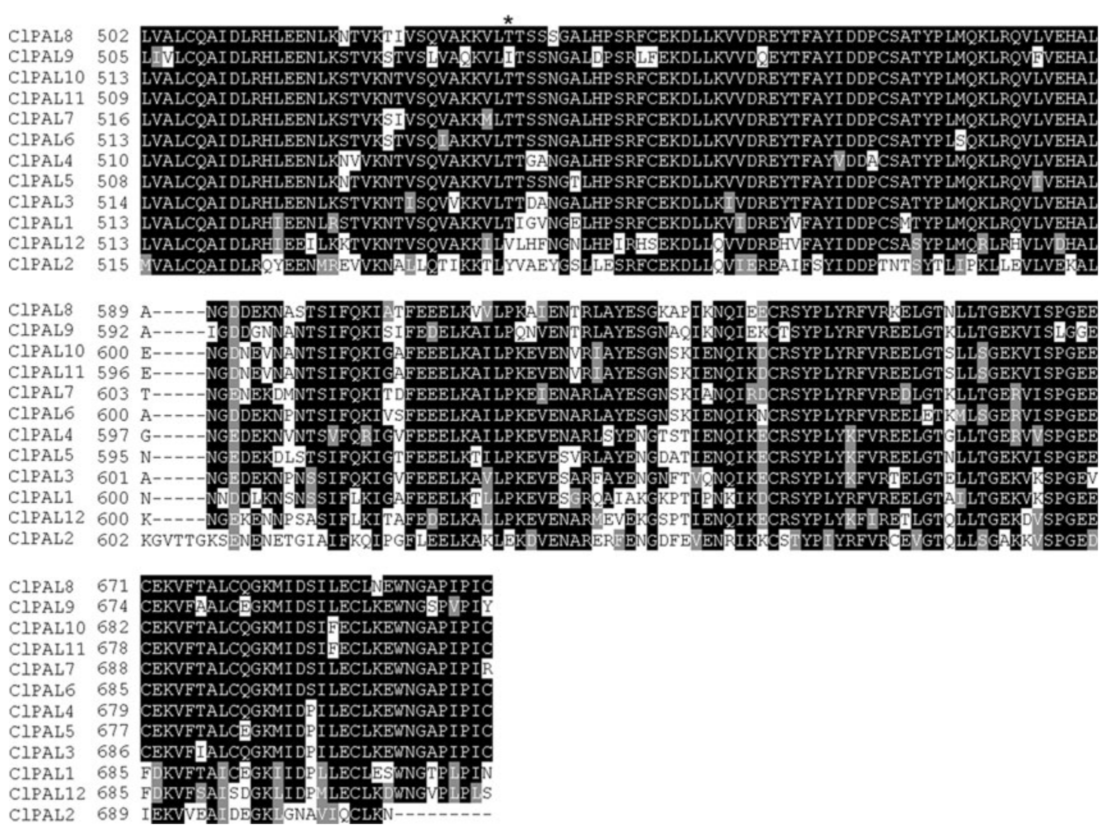


Fig. 3 continued

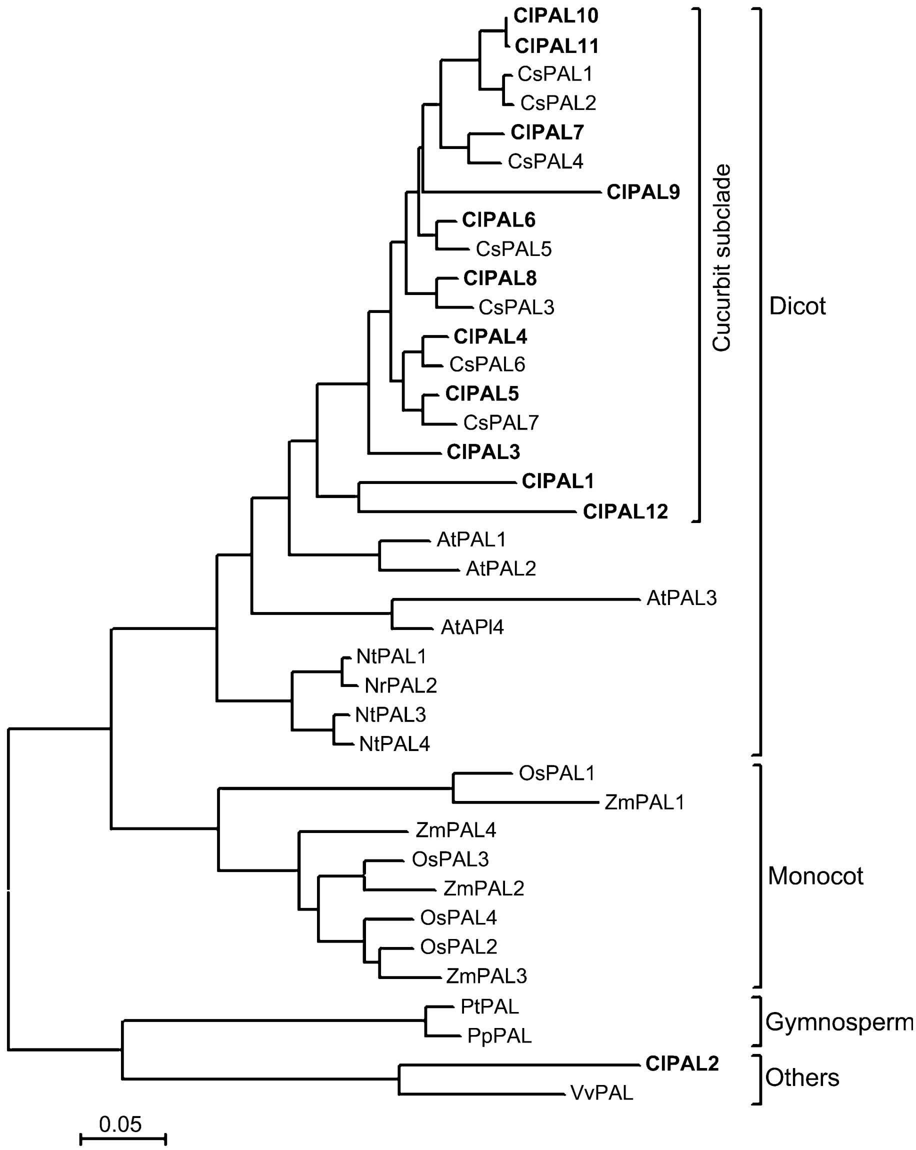


Fig. 4 Phylogenetic tree of plant PAL proteins. The conserved PAL proteins from Arabidopsis (*AtPAL*), tobacco (*NtPAL*), cucumber (*CsPAL*), watermelon (*ClPAL*), grape (*VvPAL*), rice (*OsPAL*), maize (*ZmPAL*), and pine (*PtPAL* and *PpPAL*) were aligned using Clustal X, and the phylogenic tree was constructed using the Maximum-Likelihood method with the program MEGA 5.0. The ClPAL proteins are shown in *bold*

Cis-regulatory structures of the *ClPAL* genes

It has been proposed that the regulatory sequences for most plant genes reside within 500 bp (or more) upstream of the start codon (Martin et al. 2010). To investigate the *cis*-regulatory structures of *ClPAL* promoters, approximately 2,000 bp upstream of the translational start of the *ClPAL* genes was analyzed using the PlantCARE and PLACE databases. *ClPAL10* and *ClPAL11* were found to share a common promoter with the length of only 1,709 bp because of the shorter intergenic space between these two genes (Table 1).

A total of 17 types of *cis*-regulatory elements (Table 2) were identified in the promoter regions of *ClPALs* (Fig. 6); the detailed sequence information for each *ClPAL* promoter is available in Supplementary Fig. S2. All of these *ClPAL* upstream sequences contain a putative TATA box located at the −77 to −234 bp region from the translational start codon (ATG). In addition, the popular eukaryotic regulatory element, CAAT box is also present in the promoter region of the *ClPAL* genes, with the exception of *ClPAL2* (Fig. 6). The undetectable expression of *ClPAL2* in various watermelon tissues (Fig. 5) might be attributed to the absence of a CAAT box in its promoter.

Consistent with the critical role of PAL in lignin biosynthesis (Ferrer et al. 2008), the promoters of the *ClPAL* genes, except *ClPAL2* and *ClPAL12*, contain at least one well-conserved AC-element (Fig. 6), which specifies the vascular expression of phenylpropanoid genes (Lacombe et al. 2000; Bedon et al. 2007). The presence of AC-elements drives the expression of these *ClPAL* genes in stems (Fig. 5) and suggests that the corresponding ClPAL proteins are likely involved in lignin biosynthesis in vascular lignifying cells. Additionally, an L1-box, a *cis*-element

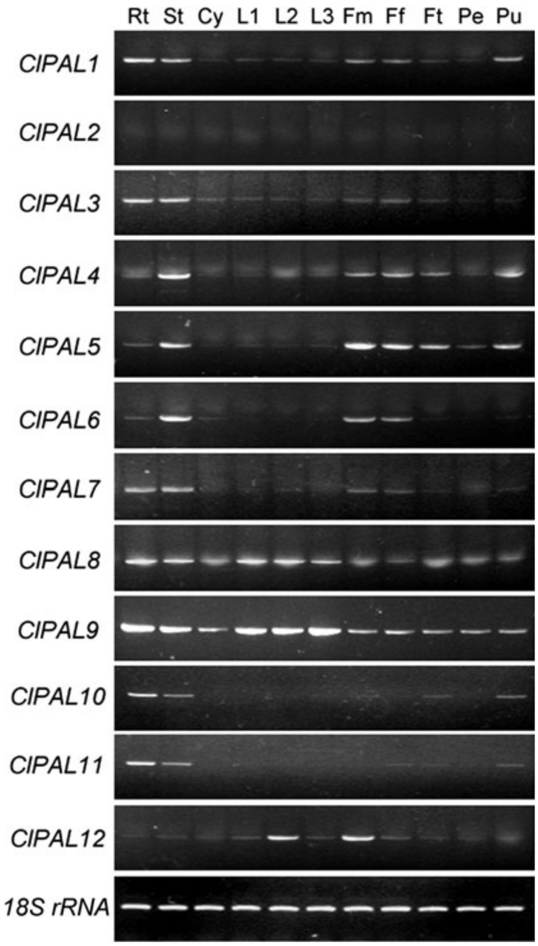


Fig. 5 Expression profiles of *ClPAL* genes in various watermelon tissues, including the roots (*Rt*), stems (*St*), cotyledons (*Cy*), mature leaves (the first leave, L1; the second leaves, L2), expanding leaves (the third leaves, L3), male (*Fm*) and female (*Ff*) flowers, and fruits (*Ft*). The fruits also contain peel (*Pe*) and pulp (*Pu*). The expression was determined by RT-PCR analysis; *18S rRNA* gene expression is shown as a constitutive control. Each figure represents one of at least three independent experiments that all gave essentially the same results

involved in the specific expression in the L1 layer, was found in the promoters *ClPAL8* and *ClPAL9*, with two copies in the *ClPAL2* promoter (Fig. 6), suggesting that ClPAL8, ClPAL9, and CsPAL2 might be involved in L1-layer development during shoot epidermal cell differentiation (Abe et al. 2001, 2003). Another potential element, Fruit-motif (Yamagata et al. 2002; Yin et al. 2009), was found in the promoters of six *ClPALs* (Fig. 6), which might drive these *ClPAL* genes to be expressed in watermelon fruit (Fig. 5).

Notably, several phytohormone-responsive elements were also found in the *ClPAL* gene promoters, including ABRE for abscisic acid (ABA) responses, AuxRE for auxin responses, ERE for ethylene responses, and GARE for gibberellins (GA) responses (Table 2; Fig. 6). In particular, seven putative GAREs were found in the promoter of *ClPAL2* (Fig. 6), indicating that *ClPAL2* expression might be the strongly induced by GA. Additionally, an SARE for salicylic acid (SA) responsiveness was identified in *ClPAL1*, *ClPAL4*, *ClPAL9*, and *ClPAL6* promoters, with two copies present in the *ClPAL3*, *ClPAL4*, and *ClPAL5* promoters (Fig. 6). The SAREs might be responsible for the expression of these *ClPAL* genes during systemic acquired resistance. Another elicitor-regulatory element, the W-box, was present in all the *ClPAL* promoters (Table 2; Fig. 6). The promoters of *ClPAL3*, *ClPAL4*, and *ClPAL5* also contained one or two elicitor-responsive elements (ELREs) (Fig. 6), and the presence of these *cis*-elements indicated pathogen- or elicitor-induced *ClPAL* gene expression (Rushton et al. 1996). In addition to the regulation by biotic elicitors, *ClPALs* might also respond to various abiotic stresses, such as dehydration, heat, and cold. An MYB-binding site involved in drought-responsiveness (Abe et al. 1997) was found in all the *ClPAL* promoters except for *ClPAL3* (Fig. 6). In the promoters of *ClPAL4*, *ClAPL8*, and *ClPAL9*, we also found the binding site for MYC, the transcription factor involved in early responsiveness to drought (Tran et al. 2004). All of the *ClPAL* promoters contain at least one heat shock-responsive element (HSE). A low-temperature-responsive element (LTRE) is also present in the promoters of six *ClPAL* genes, including *ClPAL1*, *ClPAL3*, *ClPAL5*, *ClPAL7*, *ClPAL9*, and *ClPAL12* (Fig. 6). All of the aforementioned putative *cis*-elements suggested that *CsPALs* might respond to environmental stresses via a complex mechanism and that each *ClPAL* gene would be induced by different environmental stimuli.

Discussion

The duplicated *PAL* gene family in watermelon

PAL has been reported as the first enzyme of phenylpropanoid and flavonoid biosynthesis and, therefore, plays a fundamental role in plant development and the responses to various biotic and abiotic stresses. PAL is encoded by a multigene family with four members in Arabidopsis (Raes et al. 2003) and tobacco (Reichert et al. 2009), nine members in rice (Hamberger et al. 2007), and more than 20 copies in tomato and potato (Chang et al. 2008). In the present study, we demonstrated that watermelon possesses a total of 12 *ClPAL* genes, even more than the seven *PAL* genes in another cucurbit, cucumber, as reported in our previous study (Shang et al. 2012). By comparing the genome sizes of cucumber (367 Mb; Huang et al. 2010) and Arabidopsis (125 Mb; TAIR), the size of the *PAL* gene family in watermelon (genome size 425 Mb; Guo et al. 2013) is

Table 2 Putative *cis*-element sequences in promoter regions of *ClPAL* gene family

Symbol	Cis-element	Sequence[a]	Description	Reference
T	TATA box	TATTTAA	Binding site for TBP2, homologous to the TATA box found in the promoter of rice *PAL* gene	Zhu et al. (2002)
		TATAAAT	TATA box found in pea (*Pisum sativum*); tobacco (*Nicotiana tabacum*); bean (*Phaseolus vulgaris*)	Grace et al. (2004)
C	CAAT box	CCAAT	Common *cis*-acting element in eukaryotic promoter and enhancer regions	Vauterin et al. (1999)
A	AC-element	TCCACCAACCCCYTYYM	Homologous to AC-element from the *PAL2* gene from *Phaseolus vulgaris*, conferring enhanced xylem expression and repression in phloem	Lacombe et al. (2000); Bedon et al. (2007)
B	ABRE	ACGTG	Abscisic acid (ABA)-responsive element	Nakashima et al. (2006)
X	AuxRE	KGTCCCAT	Auxin-responsive element	Ballas et al. (1993)
E	ERE	AWTTCAAA	Ethylene-responsive element, most likely involved in petal senescence	Tapia et al. (2005)
L	ELRE	TTGACC	Elicitor-responsive element	Rushton et al. (1996)
F	Fruit-motif	TCCAAAA	A fruit-specific motif identified from watermelon (*Citrullus vulgaris*)	Yin et al. (2009)
		TGTCACA	A motif involved in fruit-specific expression of the *cucumisin* genes	Yamagata et al. (2002)
G	GARE	TAACAAR	Gibberellin-responsive element	Ogawa et al. (2003)
H	HSE	CTNGAANNTTCNAG	Heat shock-responsive element	Gurley et al. (1986); Barros et al. (1992)
I	I-box	GATAAG	A *cis*-element involved in light responsiveness	Giuliano et al. (1988)
J	L1-box	TAAATGYA	A *cis*-element involved in the specific expression of L1 layer	Abe et al. (2001, 2003)
D	LTRE	RYCGAC	Low temperature-responsive element	Xue (2002)
M	MYB	CTAACCA	Binding site for MYB proteins, involved in dehydration responsiveness	Abe et al. (1997)
N	MYC	CATGTG	MYC recognition site, involved in early responsiveness to dehydration	Tran et al. (2004)
S	SARE	TCATTTTCTT	Salicylic acid responsive element, also known as TCA-element	Goldsborough et al. (1993)
W	W box	TTTGACY	WRKY protein-specific binding site, responsive to pathogen infection	Eulgem et al. (2000)
U	Wus	TTAATGG	Target sequence of WUSCHEL-type protein, involved in meristem development in both root and shoot	Kamiya et al. (2003)

[a] N—A, C, G or T; K—G or T; M—A or C; R—A or G; W—A or T; Y—C or T

proportional to that in Arabidopsis and cucumber. Recent gene duplication events are important for the rapid expansion and evolution of plant gene families (Cannon et al. 2004). The expansion of the Arabidopsis *PAL* gene family may be due to recent whole-genome duplications (WGDs), which have occurred twice in the evolution of Arabidopsis. However, in watermelon, a genome for which there is a lack of evidence for a recent WGD (Guo et al. 2013), the expansion of *PAL* genes may be attributed to recent segmental or tandem duplications, similar to the situation in cucumber (Shang et al. 2012). This conclusion can be supported by the following findings: in Arabidopsis, all four of the *PAL* isologs are widely dispersed throughout the genome, with *AtPAL1* and *AtPAL3* located on chromosomes 2 and 5, respectively, and *AtPAL2* and *AtPAL4* located on the short arm and long arm of chromosome 3, respectively (Huang et al. 2010); conversely, nine of the 12 *ClPAL* genes are arranged in tandem orientations on two chromosomal fragments (Table 1).

The phylogenetic analysis of plant PALs provides us an opportunity to investigate the evolutionary history of the plant *PAL* genes and to estimate the duplication events during the expansion of the *PAL* gene family. After the divergence of dicots from monocots, the *PAL* genes were

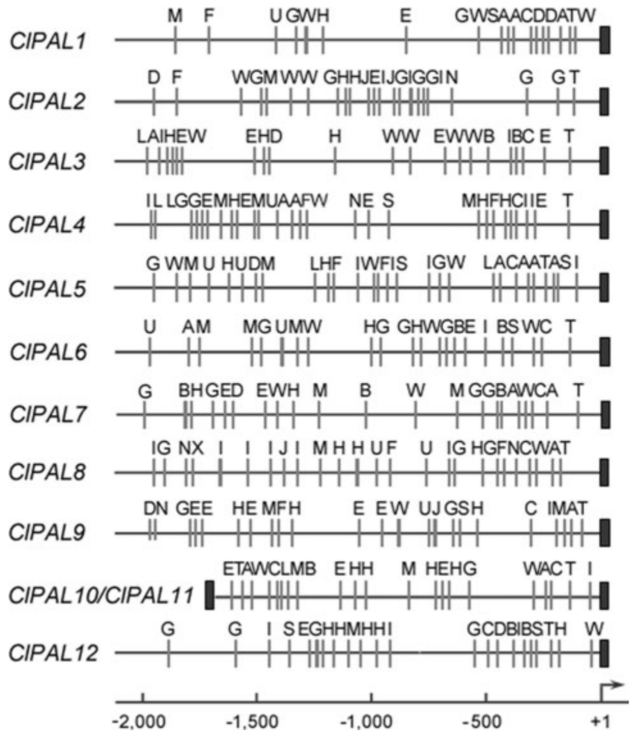


Fig. 6 Putative regulatory *cis*-elements in the *ClPAL* gene promoters. The relative positions of elements are labeled with *capital letters* in the figure and are denoted in Table 2. Detailed sequence information for each promoter and *cis*-element is available in Supplementary Fig. S2

duplicated many times, leading to the expansion of *PAL* genes in both dicots and monocots. The close paralog *PAL* genes in cucurbit, Arabidopsis, and tobacco each clustered together phylogenetically. In contrast, in the cucurbit subclade, the PALs from watermelon and cucumber were clustered together, indicating that the expansion of the watermelon PAL gene family might have occurred after the divergence of eurosids I and eurosids II (approximately 100 million years ago) (Wikström et al. 2001) but before the speciation of watermelon and cucumber that occurred 15–23 million years ago (Guo et al. 2013). The only exception was the *ClPAL10/ClPAL11* gene pair, genes that were closer to each other than to the cucumber *CsPAL1/CsPAL2* gene pair (Fig. 4), indicating that this duplication event occurred more recently, after the split of cucumber and watermelon.

It is interesting to note that the phylogenetic tree was rooted with ClPAL2 and a PAL from grape (*Vitis vinifera*, VvPAL). Grape has been considered a close relative of the eudicot ancestor (Salse 2012). Both ClPAL2 and VvPAL do not belong to the angiosperm clade but are clustered closely with the PALs from gymnosperms (Fig. 4), indicating the ancestral roles of *ClPAL2* and *VvPAL* in plant *PAL* gene evolution.

Intron insertion in the *ClPAL* genes

Parallel to the gene number, the structures of the *ClPAL* genes in watermelon have undergone evolutionary changes. Among the 12 *ClPAL* genes, nine of them (*ClPAL2* and *ClPAL4-ClPAL11*) possess no intron in their coding regions, whereas the other three *ClPAL* genes contain a single intron in their ORFs (Fig. 1). Divergent gene structures are also found in the *PAL* genes from Arabidopsis (*AtPAL1-4*): *AtPAL1* and *AtPAL2* contain a single intron each, whereas *AtPAL3* and *AtPAL4* are interrupted by two introns, with the second intron present in exon 2 (Fig. 1). Recently, Xu et al. (2012) found that structural divergence is very prevalent in duplicated genes and, in many cases, has led to the generation of functionally distinct paralogs and that structural divergence has played a more important role during the evolution of duplicated genes than nonduplicated genes.

Their gene structure implies the evolutionary relationships of the *ClPAL* genes. *ClPAL2*, the potential ancestral *PAL* gene (Fig. 4), has no intron in its coding region, indicating that the intron insertion events accompanied the expansion of the *ClPAL* gene family. In the phylogenetic tree, *PAL* genes with the same exon–intron structure were grouped in the same terminal branches. For example, nine *ClPAL* genes with no intron were classified into a separate group, which also included the intronless *PAL* genes from cucumber (Shang et al. 2012); the other two *ClPAL* genes containing one intron (*ClPAL1* and *ClPAL12*) were clustered into another group, which was closer to the other intron-containing plant *PAL*s (Fig. 4). However, the length and sequence of the introns diverged drastically among these genes, suggesting that the duplication events during the expansion of *ClPAL* gene family predated the insertion of introns (Fig. 7).

In spite of the different lengths, the intron insertion site in the *ClPAL* genes is highly conserved, and this conservation is also found among different plant species (Fig. 2), indicating a hot spot for intron insertion in plant *PAL* genes. Additionally, this insertion site is located in the fifth α-helix of the MIO domain where the active site of PAL enzymes is situated (Fig. 3; Ritter and Schulz 2004). The conserved splicing position after a long period of evolution might hint at the important roles of the MIO domain in PAL function, though some reports suggested that neither insertions nor the loss of introns can readily be linked to modifications in gene function (Javelle et al. 2011).

The functional conservation and divergence of *PAL* orthologs between cucumber and watermelon

In comparative genomics, the clustering of orthologous genes highlights the conservation and divergence of gene

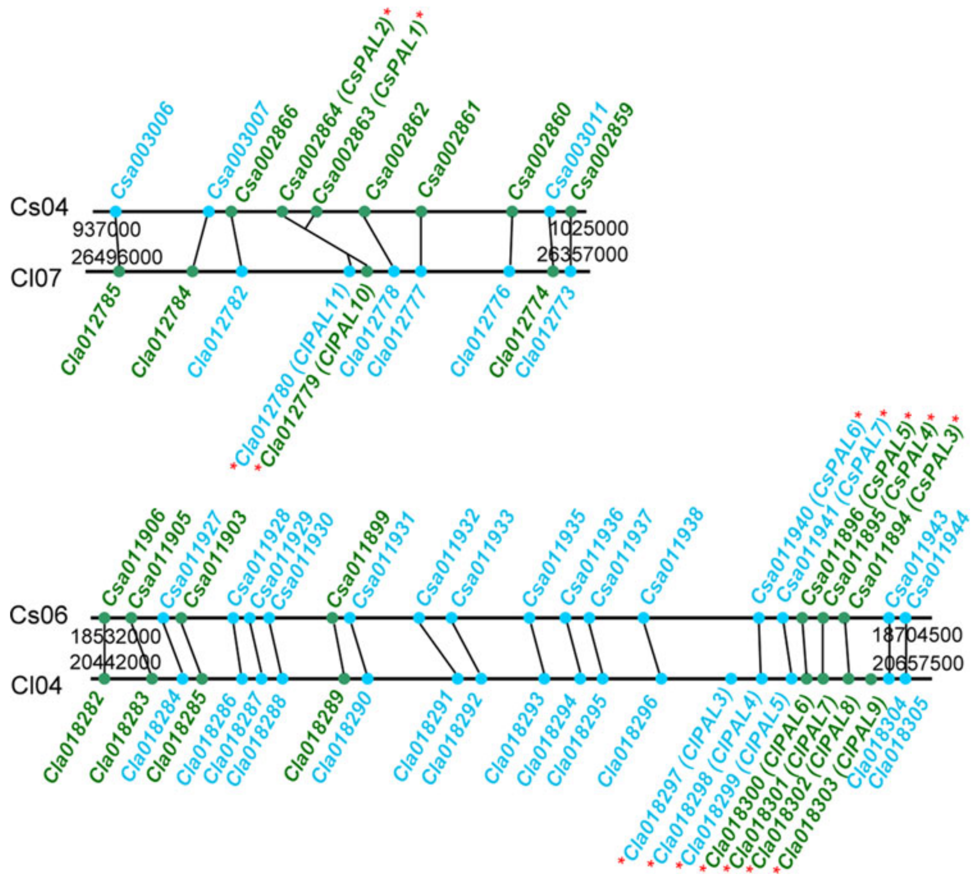


Fig. 7 Colinearity of duplicated blocks of *PAL* gene regions in cucumber and watermelon. The *blue and green dots* signify genes in the forward or reverse orientation in the relevant genomes, respectively. All mentioned *CsPAL* and *ClPAL* genes are marked with *red stars*

families among multiple genomes (Ling et al. 2011). In the present study, two strategies were used to identify orthologs or paralogs: phylogeny-based methods and colinearity analysis. Each of the tandemly duplicated *ClPAL*s (with the exception of *ClPAL3* and *ClPAL9*) was clustered with a single cucumber PAL protein; only in the case of *ClPAL10/ClPAL11* were the two watermelon genes closer to each other than to the cucumber homologs *CsPAL1/CsPAL2* (Fig. 4). To avoid the false-positive errors from phylogeny methods (Chen et al. 2007) and considering the mosaic chromosome-to-chromosome orthologous relationship between watermelon and cucumber (Huang et al. 2009; Guo et al. 2013), the genomic regions flanking the *ClPAL* gene clusters were compared with those of their cucumber counterparts. For the *ClPAL10/ClPAL11* gene cluster on chromosome 7, a genomic region on cucumber chromosome 4 was syntenic. In addition to the *PAL* gene pairs, there were eight neighboring genes colinear with the each other in the relevant genome (Fig. 7). A highly conserved gene content and order was also found between the *ClPAL3-ClPAL9* and *CsPAL3-CsPAL7* gene clusters, which were located on watermelon chromosome 4 and cucumber chromosome 6, respectively. These two syntenic blocks also contained 24 watermelon and 22 cucumber genes (Fig. 7). Last, based on the phylogeny and colinear gene arrangement, we found six orthologous pairs between watermelon and cucumber: *ClPAL4-CsPAL6*, *ClPAL5-CsPAL7*, *ClPAL6-CsPAL5*, *ClPAL7-CsPAL4*, *ClPAL8-CsPAL3*, and *ClPAL10/ClPAL11-CsPAL1/CsPAL2*.

Furthermore, correlative tissue-specific expression profiles of the orthologous *PAL* genes between watermelon and cucumber were observed. Each orthologous pair showed similar expression patterns, e. g., both *ClPAL4* and *CsPAL6* were expressed preferentially in vascular tissues and flowers. And for *ClPAL8-CsPAL3* ortholog pair, higher expression levels were detected in vegetative tissues than in productive tissues (Fig. 5; Shang et al. 2012). The correlative expression profiles of these *PAL* orthologs may be attributed to their promoters. In fact, the promoters of these *PAL* genes shared many *cis*-regulatory elements in common, such as AC-element that is for the specific expression in vascular ligninfying cells and L1-box that drive the genes expressed in L1-layers (Fig. 6; Shang et al. 2012). Comparative analysis of the *cis*-regulatory structures also hinted the correlation in their expression patterns in response to environmental stresses and plant hormones (Fig. 6; Shang et al. 2012). Also, the special distribution of these regulatory elements was conserved in each *ClPAL-CsPAL* ortholog pair (Fig. 6; Shang et al. 2012). The correlative expression profiles between *PAL* orthologs of watermelon and cucumber also facilitate us to investigate the functions of *PAL* genes from other cucurbit plants by comparative genomic approaches.

However, some of the *ClPAL* genes also showed some different expression patterns from their corresponding *CsPAL* genes. For example, some *ClPAL* genes including *ClPAL4*, *ClPAL5*, *ClPAL8*, and *ClPAL9*, were substantially expressed in the fruits, particularly in the pulp (Fig. 5), indicating a unique function of these ClPAL proteins in watermelon fruit development. Fruit color and flavor are critically dependent on the products of the phenylpropanoid pathway, and PAL, the entry point of phenylpropanoid pathway, may play a determinative role in the phenylpropanoid biosynthesis in fruit development and ripening. In fact, the roles of PAL in plant fruit development have been first reported in raspberry (*Rubus idaeus*): RiPAL1 was associated with early fruit ripening events, and RiPAL2 was correlated with the later stages of fruit development (Kumar and Ellis 2001). Whether different ClPAL isoforms participate in different stages of fruit development requires further studies. Another report on the potential roles of PAL in fruit development is from melon, where *PAL* gene expression was induced both in response to ripening and wounding (Diallinas and Kanellis 1994). Up to now, the exact fruit-specific promoting mechanisms and the related *cis*-elements remain illusive. A potential fruit-specific element identified from watermelon was TCCAAAA motif, which was located in the 5′-region of the large subunit of ADP-glucose pyrophosphorylase 1 (*AGPL1*), and could drive the reporter gene expression approximately tenfold higher in fruits than in leaves (Yin et al. 2009). Another *cis*-element involved in fruit development was TGTCACA motif, which could drive the *cucumisin* gene to be expressed in melon fruit (Yamagata et al. 2002). These two fruit-specific elements could be found as a single copy or two copies in the fruit-expressed *ClPAL* genes (Fig. 6). However, further studies are needed to determine whether these elements are really involved in the fruit-specific expression of *ClPAL* genes. Additionally, a melon *PAL* gene was found to be expressed following the kinetics of the expression of ethylene biosynthetic genes during fruit development, indicating the dependence of ethylene signals in the regulation of the fruit-specific expression of *ClPAL* genes (Diallinas and Kanellis 1994). Based on this conclusion, the putative ethylene-responsive elements (EREs) found in the *ClPAL* gene promoters may also be involved in the fruit expression of *ClPAL* genes.

As discussed earlier in this study, the expansion of *ClPAL* gene family is attributed to the segmental and tandem duplication events. The evolutionary fate of duplicated genes will be determined by their functions, and four types of functional differentiation may follow gene duplication: pseudogenization, conservation of gene function, subfunctionalization, and neofunctionalization. Subfunctionalization and neofunctionalization are the major factors for the retention of new genes, and positive selection may play an important role and during these two processes (Zhang 2003). The unique expression of the *ClPAL* genes in watermelon fruits allows the functional differentiation of these duplicates and ultimately leads to subfunctionalization or neofunctionalization. Besides, the functional differentiation of ClPALs mainly results from the differentiation of the intergenic regions. During gene duplication, the spontaneous regulon fusions or beneficial mutations can place a novel regulatory context in front of the duplicated gene candidate (Reams and Neidle 2004).

Future perspectives

The present work provides a comprehensive description of the watermelon *PAL* gene family, including the gene structure, genome localization, evolution profiles, and expression patterns, and upstream regulatory *cis*-elements of the 12 identified family members. However, some questions are still intriguing, e.g., what is the exact function for each ClPAL, and how is the expression of each *ClPAL* regulated in different developmental stages or in response to different stress or hormone signals? Thus, more molecular, biochemical, and physiological investigations are expected to further our understanding of ClPAL family. Also, the possible roles of ClPAL in watermelon fruit development may provide a potential target gene for the molecular breeding of watermelon with high-quality fruits. However, further and detailed research efforts should be made to find the detailed function of these PAL proteins and the related metabolites during watermelon fruit development and ripening. Selective silencing of the *ClPAL* genes will be informative, and for verifying the specificity of the promoter elements a transient expression assay with a promoter-reporter construct will be needed.

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